Program: **Diploma in Communication and Computer Networking**	
Course Code: **3328**	Course Title: **Operating System Design Lab**
Semester: **3**	Credits: **1.5**
Course Category: **Programme core course**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To learn Unix commands and shell programming
- To implement various CPU Scheduling Algorithms
- To implement semaphores
- To implement Deadlock Avoidance and Deadlock Detection Algorithms

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize UNIX Commands	8	Applying
CO2	Implement various CPU scheduling algorithm	10	Applying
CO3	Implement Semaphores	7	Applying
CO4	Implement Deadlock avoidance and Detection Algorithms	12	Applying
	Series Test	4	
	Open Ended Experiments / micro projects	4	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	3	2		3		
CO3	3		3				
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Familiarize UNIX Commands**		
M1.01	Basics of UNIX commands.	2	Applying
M1.02	Write programs using the following system calls of UNIX operating system fork, exec, getpid, exit, wait, close, stat, opendir, readdir	3	Applying
M1.03	Write C programs to simulate UNIX commands like cp, ls, grep, etc.	3	Applying
CO2	**Implement various CPU scheduling algorithms**		
M2.01	Write C programs to implement the various CPU scheduling algorithms.	4	Applying
M2.02	Write C program to implement threading and synchronization applications.	4	
	Series Test–I	2	
CO3	**Implement Semaphores**		
M3.01	Implementation of Semaphores**.**	2	Applying
M3.02	Implementation of the following Memory Allocation Methods for fixed partition i. First Fit ii. Worst Fit iii. Best Fit	2	Applying
M3.03	Implementation of the following Page Replacement Algorithms. i. FIFO ii. LRU iii. LFU	3	Applying
CO4	**Implement Deadlock avoidance and Detection Algorithms**		

M4.01	Bankers Algorithm for Deadlock Avoidance.	3	Applying
M4.02	Implementation of Deadlock Detection Algorithm.	3	Applying
M4.03	Implementation of the various File Organization Techniques.	3	Applying
M4.04	Implementation of the following File Allocation Strategies. Sequential Indexed Linked	3	Applying
	Series Test-II	2	
	Open ended experiment/micro projects	4	

Note: Experiments shall be conducted such that all COs are attained.
Minimum of 6 experiments (excluding open ended experiments/micro projects) shall be performed.
Compulsory for CIA
The CIA shall be arranged in third semester by the faculty in charge. The ESE need to be conducted at the end of the third semester.

Text/Reference:

T/R	Book Title/Author
T1	Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley
R1	Advanced programming in the Unix environment, W.R.Stevens, Pearson education.

Online Resources:

Sl.No	Website Link
1	http://www.tutorialspoint.com/operating_system/
2	http://courses.cs.vt.edu/~csonline/OS/Lessons/index.html
3	http://www.nptel.ac.in